

Determinants

T.T.

Q1

Apply

$$R_2 \leftarrow R_2 - 2R_1, \quad R_3 \leftarrow R_3 - R_1.$$

These row additions do not change the determinant, and they give

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & -1 & -2 \end{bmatrix}.$$

Now apply $R_3 \leftarrow R_3 + R_2$, again without changing the determinant:

$$U = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{bmatrix}.$$

There were no row exchanges or row scalings. The pivots are 1, 1, and -1 , so

$$\det A = \det U = 1 \cdot 1 \cdot (-1) = -1.$$

Q2

(i) The product rule and transpose rule give

$$\det(AB) = (\det A)(\det B) = (-2)(3) = -6, \quad \det(A^T) = \det A = -2.$$

(ii) Since $AA^{-1} = I$,

$$\det A \det(A^{-1}) = \det I = 1,$$

so

$$\det(A^{-1}) = -\frac{1}{2}.$$

Also,

$$\det(A^2) = \det A^2 = 4, \quad \det(2A) = 2^4 \det A = -32.$$

Q3

(i) The cofactors are

$$C = \begin{bmatrix} -5 & -1 & 4 \\ -1 & 2 & -8 \\ 2 & -4 & 5 \end{bmatrix}.$$

For example,

$$C_{11} = \det \begin{bmatrix} 3 & 2 \\ 4 & 1 \end{bmatrix} = -5, \quad C_{12} = -\det \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = -1.$$

The other entries follow in the same way.

(ii) Expanding along the first row gives

$$\begin{aligned} \det A &= 2C_{11} + 1C_{12} + 0C_{13} \\ &= 2(-5) + (-1) = -11. \end{aligned}$$

(iii) Using the cofactor matrix,

$$AC^T = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 2 \\ 0 & 4 & 1 \end{bmatrix} \begin{bmatrix} -5 & -1 & 2 \\ -1 & 2 & -4 \\ 4 & -8 & 5 \end{bmatrix} = \begin{bmatrix} -11 & 0 & 0 \\ 0 & -11 & 0 \\ 0 & 0 & -11 \end{bmatrix} = -11I = (\det A)I.$$

Q4

(i) Subtract the first row from the second and third rows:

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & t-1 & 1 \\ 0 & 1 & t-1 \end{bmatrix}.$$

Expanding along the first column gives

$$\det A(t) = (t-1)^2 - 1 = t(t-2).$$

(ii) The matrix is invertible exactly when its determinant is nonzero. Therefore

$$A(t) \text{ is invertible if and only if } t \neq 0 \text{ and } t \neq 2.$$

(iii) For $t = 0$, the equations $A(0)x = 0$ reduce to

$$x + y + z = 0, \quad x + 2z = 0, \quad x + 2y = 0.$$

Thus $(-2, 1, 1)$ is a nonzero null vector. For $t = 2$, the last two rows agree, and

$$(0, 1, -1)$$

is a nonzero null vector because each row has zero dot product with it.

Q5

Write the system as $Ax = b$, where

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

The determinant is

$$D = \det A = 2 \det \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - \det \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} = 2(3) - 2 = 4.$$

Replacing the first column by b gives

$$D_1 = \det \begin{bmatrix} 1 & 1 & 0 \\ 2 & 2 & 1 \\ 3 & 1 & 2 \end{bmatrix} = 2.$$

Replacing the second column gives

$$D_2 = \det \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 3 & 2 \end{bmatrix} = 0.$$

Replacing the third column gives

$$D_3 = \det \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 2 \\ 0 & 1 & 3 \end{bmatrix} = 6.$$

By Cramer's rule,

$$x = \frac{D_1}{D} = \frac{1}{2}, \quad y = \frac{D_2}{D} = 0, \quad z = \frac{D_3}{D} = \frac{3}{2}.$$

Q6

The cofactor matrix of A is

$$C = \begin{bmatrix} 0 & -2 & 2 \\ -2 & 1 & -1 \\ 2 & -1 & -3 \end{bmatrix}.$$

Expanding along the first row gives

$$\det A = 1(0) + 2(-2) + 0 = -4.$$

The cofactor formula therefore gives

$$A^{-1} = \frac{1}{\det A} C^T = -\frac{1}{4} \begin{bmatrix} 0 & -2 & 2 \\ -2 & 1 & -1 \\ 2 & -1 & -3 \end{bmatrix}.$$

The cofactor identity yields

$$AC^T = -4I,$$

and hence

$$AA^{-1} = A \left(-\frac{1}{4} C^T \right) = I.$$

Q7

(i) Put the vectors into the columns of a matrix. The parallelogram area is

$$\left| \det \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix} \right| = |12 - 2| = 10.$$

(ii) The volume is the absolute value of the determinant whose columns are u , v , and w :

$$\begin{aligned} \text{Vol} &= \left| \det \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \right| \\ &= |-2| = 2. \end{aligned}$$

Q8

- (i) The cofactor formula is

$$A^{-1} = \frac{1}{\det A} C^T.$$

Because A has integer entries, every minor and every cofactor is an integer. If $\det A = 1$ or $\det A = -1$, division by $\det A$ changes only the signs of the cofactors. Therefore every entry of A^{-1} is an integer.

- (ii) If A and A^{-1} both have integer entries, then their determinants are integers. Since

$$\det A \det(A^{-1}) = \det I = 1,$$

two integers multiply to 1. The only possibilities are

$$\det A = 1, \quad \det(A^{-1}) = 1,$$

or

$$\det A = -1, \quad \det(A^{-1}) = -1.$$

Thus $\det A = 1$ or $\det A = -1$.